

Problem 1: Constraints Elimination (0.5 pt)

Solve the following non-linear optimization problem in $\mathbb{R}^3$ with lineal equality constraints by constraint elimination:

$$\begin{aligned} &\text{minimize } f(x) = \exp(\mathbf{1}^T x) + \exp(-\mathbf{1}^T x) \\ &\text{subject to: } \begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{pmatrix} x = \begin{pmatrix} 0 \\ 5 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} &\Rightarrow \min_z f(Fz + x_0) = \tilde{f}(z) \\ &F \in \mathbb{R}^{3 \times 1} : \text{basis of } N(A) \\ &x_0 \in \mathbb{R}^3 : \text{any } x \text{ satisfying } Ax = b. \end{aligned}$$

Hint: use Gaussian elimination or QR decomposition. You are allowed to use a computer algebra system to verify your computations.

Gaussian elimination

$$\begin{aligned} &\begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 2 & 1 & 0 & | & 5 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 0 & 5 & 0 & | & 5 \end{pmatrix} \Rightarrow x_1 = 2, \text{ choose } x_3 = 0 \Rightarrow x_0 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \\ &\Rightarrow \min_z f\left(\begin{bmatrix} 0 \\ 5 \\ z \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) = e^{3+z} + e^{-3-z}, \quad \tilde{f}'(z) = e^{3+z} - e^{-3-z} \stackrel{!}{=} 0 \Rightarrow 3+z = -3-z \Rightarrow z^* = -3 \end{aligned}$$

$$N(A) = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} = Fz \Rightarrow F = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow x^* = Fz + x_0 = \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$